

COLOURING THE p -ADIC PLANE

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ABSTRACT. Let $\chi(\mathbb{Q}_p^2)$ be the chromatic number of the graph on $\mathbb{Q}_p^2$ in which (x, y) and (x', y') are adjacent when $(x - x')^2 + (y - y')^2 = 1$. Bardestani and Mallahi-Karai showed that its Borel chromatic number is finite exactly when $p = 2$ or $p \equiv 3 \pmod{4}$, and asked whether it is bounded independently of p . It is not: for $p \equiv 3 \pmod{4}$ every proper colouring with Haar-measurable colour classes uses at least $1 + (p + 1)/(2\sqrt{p})$ colours. The proof is the Delsarte–Hoffman bound on the compact group $\mathbb{Z}_p^2$, where the Fourier coefficients of the unit circle are Kloosterman sums at the first level and at most $2/(p + 1)$ at every deeper level. We also show $\chi(\mathbb{Q}_7^2) = 4$, for ordinary and for Borel colourings, while $\chi(\mathbb{Q}_2^2) = 2$ and $\chi(\mathbb{Q}_3^2) = 3$ follow from results of Madore. A theorem of Davies shows that the graph of every non-degenerate isotropic form over a field of characteristic 0 has infinite chromatic number, so the dichotomy of Bardestani and Mallahi-Karai holds for ordinary colourings too; in particular $\chi(\mathbb{Q}_p^2) = \infty$ for $p \equiv 1 \pmod{4}$. Finally, unit-distance graphs over 27 real quadratic fields give $\chi(\mathbb{Q}_p^2) \geq 4$ for every prime $p \equiv 3 \pmod{4}$ with $7 \leq p < 2\,129\,503\,819$, and a Chebotarev argument shows that no finite set of number fields gives this bound for every such p . In every dimension n we determine when $\chi(\mathbb{Q}_p^n)$ is finite, and show $\chi(\mathbb{Q}_2^4) = 4$ through the residue field $\mathbb{F}_4$ of the 2-adic quaternions.

1. INTRODUCTION

For a field F let $G(F)$ be the graph on F^2 in which v and w are adjacent when $Q(v - w) = 1$, with $Q(x, y) = x^2 + y^2$, and write $\chi(F^2)$ for its chromatic number. For $F = \mathbb{R}$ this is the Hadwiger–Nelson problem, and $5 \leq \chi(\mathbb{R}^2) \leq 7$: the lower bound is due to de Grey [7] and to Exoo and Ismailescu [8], the upper bound to Isbell (see [21]). Woodall showed $\chi(\mathbb{Q}^2) = 2$ [25], and the planes over number fields have been studied by Johnson [12], Fischer [9], Madore [13], Moorhouse [17] and others. Madore observed that reduction modulo a prime gives $\chi(\mathbb{Q}_p^2) \leq \chi(\mathbb{F}_p^2)$ for $p \equiv 3 \pmod{4}$ [13, ¶1.5 and Remark 3.6].

Bardestani and Mallahi-Karai [3] studied the graphs of quadratic forms over local fields. For a local field F of characteristic 0 they proved that the *Borel* chromatic number of the graph of a non-degenerate form is infinite if and only if the form is isotropic, using the spectral bound of Bachoc, DeCorte, Oliveira and Vallentin [2]. Over $\mathbb{Q}_p$ the form $x^2 + y^2$ is anisotropic exactly when -1 is not a square, that is for $p = 2$ and $p \equiv 3 \pmod{4}$. For $p \equiv 3 \pmod{4}$ they noted that the Borel chromatic number $\chi_B(\mathbb{Q}_p^2)$ is $O(p^2)$, and asked:

Question ([3, Question 1], the p -adic Hadwiger–Nelson problem). Is there a constant B with $\chi_B(\mathbb{Q}_p^2) \leq B$ for every prime $p \equiv 3 \pmod{4}$?

Write χ_m for the measurable chromatic number (colour classes measurable for the Haar measure), so that $\chi \leq \chi_m \leq \chi_B$. Our first result answers the question in the negative. We have not found another answer in the literature, including the later papers of the same authors [4, 5].

Theorem 1. *For every prime $p \equiv 3 \pmod{4}$,*

$$1 + \frac{p + 1}{2\sqrt{p}} \leq \chi_m(\mathbb{Q}_p^2) \leq \chi_B(\mathbb{Q}_p^2) \leq \chi(\mathbb{F}_p^2),$$

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and $\chi(\mathbb{F}_p^2) \leq (p+1)/2$ for $p > 3$. In particular $\chi_B(\mathbb{Q}_p^2) \rightarrow \infty$.

Here $\mathbb{F}_p^2$ denotes the graph $G(\mathbb{F}_p)$. Le Anh Vinh [22, Theorem 1] showed $(\frac{1}{2} + o(1))\sqrt{q} \leq \chi(\mathbb{F}_q^2) \leq (q + q/r)/2$ for powers $q > 3$ of an odd prime r ; the upper bound in Theorem 1 is his. The lower bound of Theorem 1 has the same leading term as his, but it holds for all measurable colourings of the p -adic plane, not only for those that factor through $\mathbb{F}_p^2$. The exact Fourier coefficients give better values for small p (Table 1): at least 5 colours from $p = 23$, 6 for $p = 59$ and every $p \geq 67$, 7 for $p = 71$ and every $p \geq 103$.

For the ordinary chromatic number:

Theorem 2. $\chi(\mathbb{Q}_7^2) = \chi_B(\mathbb{Q}_7^2) = 4$. Moreover $\chi(\mathbb{Q}_2^2) = \chi_B(\mathbb{Q}_2^2) = 2$ and $\chi(\mathbb{Q}_3^2) = \chi_B(\mathbb{Q}_3^2) = 3$.

The upper bounds are Madore's reductions [13, Propositions 3.2 and 3.8], which give locally constant colourings, and the values 2 and 3 follow at once from his results (his 9-cycle over $\mathbb{Q}(\sqrt{7})$ [13, Proposition 4.3] lies in $\mathbb{Q}_3^2$). The new value is $\chi(\mathbb{Q}_7^2) = 4$. Since 3 is not a square modulo 7, the plane $\mathbb{Q}_7^2$ contains no unit equilateral triangle (the third vertex of a unit triangle on 0 and (a, b) is $(\frac{a}{2} - \frac{\sqrt{3}}{2}b, \frac{b}{2} + \frac{\sqrt{3}}{2}a)$, which forces $\sqrt{3}$ into the field), so the classical 4-chromatic graphs, built from triangles, do not embed; we use a triangle-free 76-vertex graph over $\mathbb{Q}(\sqrt{11})$ with no proper 3-colouring [11], as $11 \equiv 2^2 \pmod{7}$. The three ordinary values are also proved in Lean 4 with Mathlib [15, 14], in `lean/PadicPlanes.lean` of [10]; for instance `(QuadraticPlanes.sumSqGraph Q_7).chromaticNumber = 4`.

In the isotropic case the ordinary chromatic number is infinite as well, by a theorem of Davies [6]:

Proposition 3. *Let F be a field of characteristic 0 and Q a non-degenerate isotropic quadratic form on F^n , $n \geq 2$. Then the graph on F^n in which v, w are adjacent when $Q(v - w) = 1$ has infinite chromatic number. In particular $\chi(F^2) = \infty$ when -1 is a square in F , and $\chi(\mathbb{Q}_p^2) = \infty$ for $p \equiv 1 \pmod{4}$.*

Corollary 4. *Let F be a local field of characteristic 0 and Q a non-degenerate quadratic form in at least two variables over F . Then the chromatic number of its graph is finite if and only if Q is anisotropic, if and only if its Borel chromatic number is finite. In particular $\chi(\mathbb{Q}_p^2)$ is finite exactly when $p = 2$ or $p \equiv 3 \pmod{4}$.*

Proof. If Q is anisotropic, [3, Theorem 1.1] gives $\chi \leq \chi_B < \infty$; if it is isotropic, Proposition 3 gives $\chi_B \geq \chi = \infty$. $\square$

In all dimensions (Proposition 11 in §6), $\chi(\mathbb{Q}_p^n)$ is finite exactly when $n = 1$, or $n = 2$ and $p \not\equiv 1 \pmod{4}$, or $p = 2$ and $n \leq 4$; and $\chi(\mathbb{Q}_2^2) = \chi(\mathbb{Q}_3^2) = 2$, $\chi(\mathbb{Q}_2^4) = 4$. On the p -adic line, $\chi(\mathbb{Q}_p^1) = 2$ but $\chi_B(\mathbb{Q}_p^1) = 3$ for odd p .

Finally, every number field K with an embedding $K \hookrightarrow \mathbb{Q}_p$ gives $\chi(\mathbb{Q}_p^2) \geq \chi(K^2)$. Unit-distance graphs with no proper 3-colouring over 27 real quadratic fields [11] give:

Theorem 5. $\chi(\mathbb{Q}_p^2) \geq 4$ for every prime $p \equiv 3 \pmod{4}$ with $7 \leq p < 2\,129\,503\,819$. Moreover $\chi(\mathbb{Q}_{83}^2) \geq 5$.

No finite set of number fields can do this for every p :

Proposition 6. *Let $\mathcal{K}$ be a finite set of number fields with $\chi(K^2) \geq 4$ for every $K \in \mathcal{K}$. Then there are infinitely many primes $p \equiv 3 \pmod{4}$ such that no $K \in \mathcal{K}$ embeds in $\mathbb{Q}_p$.*

For the 27 fields of Theorem 5 the first such prime after 3 is 2 129 503 819. The proposition is about graphs over fixed fields; a single abstract graph might be realised in $\mathbb{Q}_p^2$ over different fields for different p .

2. REDUCTION MODULO p : UPPER BOUNDS

Lemma 7 (Madore [13, Propositions 3.2 and 3.8]). *Let $p = 2$ or $p \equiv 3 \pmod{4}$. If $x, y \in \mathbb{Q}_p$ and $x^2 + y^2 = 1$, then $x, y \in \mathbb{Z}_p$, and $(x, y) \pmod{p}$ is a nonzero point of the circle $C_p = \{(a, b) \in \mathbb{F}_p^2 : a^2 + b^2 = 1\}$.*

Proof. Suppose $-k = \min(v_p(x), v_p(y)) < 0$. If only one of x, y had valuation $-k$, then $v_p(x^2 + y^2) = -2k \neq 0$. So $x = p^{-k}a$, $y = p^{-k}b$ with $a, b \in \mathbb{Z}_p^\times$ and $a^2 + b^2 = p^{2k}$. For odd p this gives $(a/b)^2 \equiv -1 \pmod{p}$, impossible for $p \equiv 3 \pmod{4}$. For $p = 2$, odd squares are 1 mod 8, so $v_2(a^2 + b^2) = 1$, while $v_2(p^{2k})$ is even. Hence $x, y \in \mathbb{Z}_p$, and reducing $x^2 + y^2 = 1$ gives the rest. $\square$

Proposition 8 (cf. [13, Remark 3.6]). *If $p = 2$ or $p \equiv 3 \pmod{4}$, then $\chi_B(\mathbb{Q}_p^2) \leq \chi(\mathbb{F}_p^2)$, by a locally constant colouring.*

Proof. Let $r(v)$ be the representative of $v + \mathbb{Z}_p^2$ whose coordinates are sums $\sum_{j < 0} a_j p^j$ with $0 \leq a_j < p$; it is locally constant. Given a proper colouring κ of $\mathbb{F}_p^2$, put $c(v) = \kappa((v - r(v)) \bmod p)$. If v and w are adjacent, $v - w \in \mathbb{Z}_p^2$ by Lemma 7, so $r(v) = r(w)$, and $(v - r(v)) - (w - r(w)) = v - w$ reduces to a point of C_p ; hence $c(v) \neq c(w)$. The colouring is locally constant, so it is Borel. $\square$

For $p = 2$, $C_2 = \{(1, 0), (0, 1)\}$ and $\kappa(a, b) = a + b$ is a proper 2-colouring; for $p = 3$, $C_3 = \{(\pm 1, 0), (0, \pm 1)\}$ and $\kappa(a, b) = a + b \bmod 3$ is a proper 3-colouring. For $p = 7$ the graph $\mathbb{F}_7^2$ is 4-colourable (Moorhouse [17], Vinh [22]; a colouring is stored in [10]).

3. MEASURABLE COLOURINGS: PROOF OF THEOREM 1

The following lemma is the ratio bound of Bachoc, DeCorte, Oliveira and Vallentin [2], which [3, Theorem 2.5] states on $\mathbb{Q}_p^n$; we include the short proof for completeness. The new input is Lemma 10.

Lemma 9 (Delsarte–Hoffman on a compact group). *Let Γ be a compact abelian group with Haar probability measure μ , and σ a probability measure on a closed set $S \subset \Gamma$ with $S = -S$, $0 \notin S$ and σ invariant under $s \mapsto -s$. For a character ψ of Γ put $\hat{\sigma}(\psi) = \int \psi d\sigma$, which is real, and let $m = \inf\{\hat{\sigma}(\psi) : \psi \neq 1\}$, the infimum over the nontrivial characters; suppose $m < 0$. If $A \subset \Gamma$ is measurable and $A \cap (A + s) = \emptyset$ for every $s \in S$, then $\mu(A) \leq -m/(1 - m)$.*

Proof. The function $f(x) = \mu(A \cap (A - x))$ is continuous and positive definite, with $\hat{f}(\psi) = |\hat{1}_A(\psi)|^2 \geq 0$, $\hat{f}(1) = \mu(A)^2$ and $\sum_\psi \hat{f}(\psi) = \|1_A\|_2^2 = \mu(A)$ (Parseval); so $f = \sum_\psi \hat{f}(\psi)\psi$ uniformly. By hypothesis $f = 0$ on S , hence

$$0 = \int f d\sigma = \sum_\psi \hat{f}(\psi) \hat{\sigma}(\psi) \geq \mu(A)^2 + m(\mu(A) - \mu(A)^2),$$

which gives the bound. $\square$

Let $p \equiv 3 \pmod{4}$. Identify $\mathbb{Z}_p^2$ with the ring $\mathcal{O} = \mathbb{Z}_p[i]$ of integers of the unramified quadratic extension $E = \mathbb{Q}_p(i)$ by $(x, y) \mapsto x + iy$; then Q becomes the norm $Nz = z\bar{z}$, and the unit circle is the compact group $S = \{z \in \mathcal{O} : Nz = 1\}$, with its Haar probability measure σ . Since $E/\mathbb{Q}_p$ is unramified and p is odd, the norm maps $\mathcal{O}^\times$ onto $\mathbb{Z}_p^\times$ and $1 + p^r \mathcal{O}$ onto $1 + p^r \mathbb{Z}_p$ for every $r \geq 1$; so the image of S in $(\mathcal{O}/p^r)^\times$ is the whole norm-one subgroup, of order $(p + 1)p^{r-1}$, and the classes of S modulo p^r all have measure $1/((p + 1)p^{r-1})$. Write $e(t) = \exp(2\pi it)$ and $e_p(t) = e(\{t\}_p)$ for $t \in \mathbb{Q}_p$. The characters of $\mathcal{O}$ are $\psi_\xi(z) = e_p(\text{Tr}(\xi z))$ for $\xi \in E/\mathcal{O}$ (the different of $E/\mathbb{Q}_p$ is trivial), the trivial one being $\xi = 0$; call $\ell = -v_p(\xi)$ the *level* of ψ_ξ . Since $-1 \in S$ and $\bar{S} = S$, every $\hat{\sigma}(\xi) = \int_S \psi_\xi d\sigma$ is real.

Lemma 10. *Let $\xi = p^{-\ell}\xi_0$ with $\xi_0 \in \mathcal{O}^\times$.*

- (1) *If $\ell = 1$, then $\hat{\sigma}(\xi) = \lambda/(p + 1)$, where λ is an eigenvalue of the graph $\mathbb{F}_p^2$; more precisely $\lambda = -K(n)$ with $n = N\xi_0 \bmod p$ and the Kloosterman sum $K(n) = \sum_{x \in \mathbb{F}_p^\times} e((x + n/x)/p)$, so $|\lambda| \leq 2\sqrt{p}$.*
- (2) *If $\ell \geq 2$, then $|\hat{\sigma}(\xi)| \leq 2/((p + 1)p^{\lfloor \ell/2 \rfloor - 1}) \leq 2/(p + 1)$.*

Proof. (1) The character factors through $\mathcal{O}/p\mathcal{O} = \mathbb{F}_{p^2}$, and S maps onto the norm-one group $\bar{C} = \{z \in \mathbb{F}_{p^2} : z\bar{z} = 1\}$, of order $p + 1$, with fibres of equal measure. So $\hat{\sigma}(\xi) = \frac{1}{p+1} \sum_{z \in \bar{C}} \psi(\text{Tr}(cz))$ with $c = \xi_0 \bmod p$ and $\psi(t) = e(t/p)$, the eigenvalue of $\mathbb{F}_p^2$ at the corresponding character. That

these eigenvalues are Kloosterman sums is due to Medrano, Myers, Stark and Terras [16]; here is the short argument. With $n = Nc \neq 0$, $w = cz$ gives $\sum_{z \in \bar{C}} \psi(\text{Tr}(cz)) = \sum_{Nw=n} \psi(\text{Tr } w)$. An element w with $Nw = n$ and $\text{Tr } w = t$ is a root of $X^2 - tX + n$, and conversely each root outside $\mathbb{F}_p$ has this norm and trace, while a root $w \in \mathbb{F}_p$ has norm w^2 and trace $2w$, which fits only when $t^2 = 4n$. So there are $1 - \eta(t^2 - 4n)$ such w , where η is the quadratic character of $\mathbb{F}_p$. Similarly $x + n/x = t$ has $1 + \eta(t^2 - 4n)$ solutions $x \in \mathbb{F}_p^\times$. As $\sum_t \psi(t) = 0$, the first sum is $-\sum_t \psi(t)\eta(t^2 - 4n) = -K(n)$, and Weil's bound [24] gives $|K(n)| \leq 2\sqrt{p}$.

(2) Let $j = \lceil \ell/2 \rceil$, $r = \ell - j = \lfloor \ell/2 \rfloor \geq 1$, and $S_j = \{s \in S : s \equiv 1 \pmod{p^j}\}$. For $s_0 \in S$, the function $s \mapsto \psi_\xi(s_0 s) / \psi_\xi(s_0) = e_p(\text{Tr}(\xi s_0(s - 1)))$ is a character of S_j , because $(1 + p^j t)(1 + p^j t') = 1 + p^j(t + t') + p^{2j}tt'$ and $v_p(\xi p^{2j}) \geq 0$. Its average over S_j is 0 unless it is trivial. The elements $(1 + p^j ia/2)/(1 - p^j ia/2)$, $a \in \mathbb{Z}_p$, lie in S_j and are $\equiv 1 + p^j ia \pmod{p^{2j}}$, so triviality forces $e_p(\text{Tr}(i p^{j-\ell} a \xi_0 s_0)) = 1$ for all $a \in \mathbb{Z}_p$, that is, $\text{Tr}(i \xi_0 s_0) = -2 \text{Im}(\xi_0 s_0) \equiv 0 \pmod{p^r}$. Then $\xi_0 s_0 \equiv u \pmod{p^r}$ for some $u \in \mathbb{Z}_p$ with $u^2 \equiv N \xi_0 \pmod{p^r}$, so s_0 lies in one of at most two classes modulo p^r , each of measure $1/((p+1)p^{r-1})$; and the cosets of S_j refine these classes since $j \geq r$. Averaging over the cosets of S_j first gives the bound. $\square$

Proof of Theorem 1. The upper bounds are Proposition 8 and [22, Theorem 1]. For the lower bound, let c be a proper colouring of $\mathbb{Q}_p^2$ with k measurable classes. Their traces on $\mathbb{Z}_p^2$ cover it, so one of them, A , has $\mu(A) \geq 1/k$, and since $S \subset \mathbb{Z}_p^2$ no two points of A differ by an element of S . By Lemma 10, $m \geq -\max(2\sqrt{p}, 2)/(p+1) = -2\sqrt{p}/(p+1)$, and $m < 0$ because the eigenvalues of $\mathbb{F}_p^2$ sum to 0; so Lemma 9 gives $1/k \leq \mu(A) \leq |m|/(1+|m|)$, that is, $k \geq 1 + 1/|m| \geq 1 + (p+1)/(2\sqrt{p})$. $\square$

With the exact least eigenvalue $\lambda_{\min}$ of $\mathbb{F}_p^2$, the same proof gives $\chi_m(\mathbb{Q}_p^2) \geq 1 + (p+1)/\max(|\lambda_{\min}|, 2)$. We computed $\lambda_{\min}$ in interval arithmetic (`padic_measurable.py` in [10]); Table 1 shows the result. Weil's bound alone gives at least 6 colours for $p \geq 62$, 7 for $p \geq 98$, 8 for $p \geq 142$ and 9 for $p \geq 194$; with the exact spectra (computed for $p < 200$), $\chi_m(\mathbb{Q}_p^2) \geq 6$ already for $p = 59$, ≥ 7 for $p = 71$, ≥ 8 for every $p \geq 131$, and ≥ 9 for $p = 191$.

p	$\lambda_{\min}$	$1 + 1/ m $	$\chi_m(\mathbb{Q}_p^2) \geq$
3	-2.0000	3.0000	3
7	-4.4940	2.7802	3
11	-4.7958	3.5022	4
19	-7.6097	3.6282	4
23	-7.9607	4.0148	5
31	-10.6665	4.000044	5
43	-11.9622	4.6782	5
47	-12.8465	4.7364	5
59	-14.9176	5.0221	6
67	-15.0567	5.5163	6
71	-14.1728	6.0801	7
79	-17.1019	5.6778	6
83	-17.7087	5.7434	6
103	-20.0322	6.1916	7
107	-19.0495	6.6694	7

TABLE 1. Lower bounds for the measurable chromatic number of $\mathbb{Q}_p^2$, $p \equiv 3 \pmod{4}$, from the least eigenvalue of $\mathbb{F}_p^2$ (rounded; the bounds are rigorous lower ends of 120-bit intervals).

Together with Theorem 2, $\chi_m(\mathbb{Q}_p^2) = \chi_B(\mathbb{Q}_p^2) = \chi(\mathbb{Q}_p^2)$ for $p = 2, 3, 7$, while $4 \leq \chi_m(\mathbb{Q}_p^2) \leq \chi_B(\mathbb{Q}_p^2) \leq 5$ for $p = 11, 19$, where $\chi(\mathbb{F}_p^2) = 5$ ([10] stores 5-colourings of $\mathbb{F}_{11}^2$ and $\mathbb{F}_{19}^2$, and certificates that 4 colours do not suffice).

4. LOWER BOUNDS FROM NUMBER FIELDS

If K is a field and $\iota : K \hookrightarrow \mathbb{Q}_p$, then $(x, y) \mapsto (\iota x, \iota y)$ is an injective homomorphism $G(K) \rightarrow G(\mathbb{Q}_p)$, so $\chi(\mathbb{Q}_p^2) \geq \chi(K^2)$. For odd p and $p \nmid d$, Hensel's lemma gives $\mathbb{Q}(\sqrt{d}) \subset \mathbb{Q}_p$ exactly when d is a square modulo p .

Proof of Theorem 2. The upper bounds are Proposition 8. The graph has edges, so $\chi(\mathbb{Q}_2^2) \geq 2$. For $p = 3$, besides Madore's 9-cycle, the unit vectors

$$(1, 0), \quad (0, 1), \quad \left(\frac{-1+\sqrt{7}}{4}, \frac{-1-\sqrt{7}}{4}\right), \quad \left(-\frac{\sqrt{7}}{4}, -\frac{3}{4}\right), \quad \left(-\frac{3}{4}, \frac{\sqrt{7}}{4}\right)$$

sum to 0, and their partial sums $0, (1, 0), (1, 1), (\frac{3+\sqrt{7}}{4}, \frac{3-\sqrt{7}}{4}), (\frac{3}{4}, -\frac{\sqrt{7}}{4})$ are distinct, so $G(\mathbb{Q}(\sqrt{7}))$ contains a 5-cycle; as $7 \equiv 1 \pmod{3}$, $\sqrt{7} \in \mathbb{Q}_3$ and $\chi(\mathbb{Q}_3^2) \geq 3$. Finally $11 \equiv 2^2 \pmod{7}$, so $\sqrt{11} \in \mathbb{Q}_7$, and the 76-vertex unit-distance graph over $\mathbb{Q}(\sqrt{11})$ of [11] has no proper 3-colouring (a DRAT proof checked by `drat-trim` and by the verified checker `cake_lpr`, and a proof in Lean); so $\chi(\mathbb{Q}_7^2) \geq 4$. $\square$

Proof of Theorem 5. Let

$$\mathcal{D} = \{11, 23, 35, 47, 59, 71, 95, 119, 131, 155, 179, 191, 239, 251, \\ 263, 359, 431, 443, 455, 491, 599, 611, 791, 851, 911, 935, 959\}.$$

For each $d \in \mathcal{D}$, [11] gives a unit-distance graph over $\mathbb{Q}(\sqrt{d})$ with no proper 3-colouring. A segmented sieve with Jacobi symbols (`padic_reach.c` in [10], about half a minute) shows that for every prime $p \equiv 3 \pmod{4}$ with $7 \leq p < 2\,129\,503\,819$ some $d \in \mathcal{D}$ is a nonzero square modulo p , so $\mathbb{Q}(\sqrt{d}) \subset \mathbb{Q}_p$ and $\chi(\mathbb{Q}_p^2) \geq 4$; a separate script checks every prime below 10^7 again with Euler's criterion. The prime $2\,129\,503\,819$ is the first after 3 at which every $d \in \mathcal{D}$ is a non-square. For $p = 83$, the numbers 3, 11 and 247 are squares modulo 83, and [10] contains a 5-chromatic unit-distance graph over $\mathbb{Q}(\sqrt{3}, \sqrt{11}, \sqrt{247})$. $\square$

Proof of Proposition 6. Let M be the Galois closure of the compositum of $\mathbb{Q}(i)$ and the fields in $\mathcal{K}$, $G = \text{Gal}(M/\mathbb{Q})$, $D \subset G$ the decomposition group of a prime of M above 3, $I \subset D$ its inertia group, and $\sigma \in D$ an element mapping to the Frobenius generator of D/I , so that $D = \langle \sigma \rangle I$. By Chebotarev's theorem infinitely many primes p , unramified in M , have Frobenius conjugate to σ . Since 3 is inert in $\mathbb{Q}(i)$, σ acts on the residue field $\mathbb{F}_9$ of $\mathbb{Q}(i)$ at 3 by $x \mapsto x^3$ and so moves i ; hence $p \equiv 3 \pmod{4}$. Suppose $K = M^H \in \mathcal{K}$ embeds in $\mathbb{Q}_p$. Then a prime of K above p has residue degree 1, so the Frobenius at p , and with it σ , which has the same cycle type on G/H , fixes a coset gH ; and $D \cdot gH = I \langle \sigma \rangle \cdot gH = I \cdot gH$. The primes of K above 3 correspond to the D -orbits on G/H , with residue degree the number of I -orbits in the orbit (see [19, Chapter I, §9]); so the prime given by $D \cdot gH$ has residue field $\mathbb{F}_3$, in which -1 is not a square. The argument of Lemma 7 and Proposition 8, with a uniformiser in place of p and $\kappa(a, b) = a + b \pmod{3}$, then gives $\chi(K^2) \leq 3$, a contradiction. $\square$

For quadratic fields this is Fischer's theorem that $\chi(\mathbb{Q}(\sqrt{N})^2) \leq 3$ when $N \equiv 0, 1 \pmod{3}$ [9]. A uniform bound $\chi(\mathbb{Q}_p^2) \geq 4$ for all $p \geq 7$ would follow, for example, from $\chi(\mathbb{Q}(\sqrt{q})^2) \geq 4$ for every prime $q \equiv 11 \pmod{12}$: by Dirichlet's theorem every prime $p \equiv 3 \pmod{4}$ has such a q with $q \equiv 1 \pmod{p}$, so that $\mathbb{Q}(\sqrt{q}) \subset \mathbb{Q}_p$.

5. THE ISOTROPIC CASE

Proof of Proposition 3. Let $HG(F)$ be the graph on F^2 in which (x, y) and (x', y') are adjacent when $(x - x')(y - y') = 1$. By [3, Lemma 2.7], $HG(F)$ is an induced subgraph of the graph of any isotropic form over F , and it contains $HG(\mathbb{Q})$. The map $(x, y) \mapsto (x - y, x + y)$ sends the graph of $x^2 - y^2$ on $\mathbb{Q}^2$ onto $HG(\mathbb{Q})$, because $(\Delta x)^2 - (\Delta y)^2 = (\Delta x - \Delta y)(\Delta x + \Delta y)$. Davies [6, Theorem 3], from a theorem of Graham, proved that every finite colouring of $\mathbb{Q}^2$ has a monochromatic pair with $(x - x')^2 - (y - y')^2 = 1$. Hence no finite colouring is proper. The form $x^2 + y^2$ is isotropic over F exactly when -1 is a square in F , and over $\mathbb{Q}_p$ this happens exactly when $p \equiv 1 \pmod{4}$. $\square$

Madore had observed that $\chi(\mathbb{R}^{1,1}) = \infty$ would give $\chi(\mathbb{C}^2) = \infty$ [13, ¶6.4], and Davies recorded this consequence of his theorem; the proposition is the same remark for any field of characteristic 0.

6. LINES AND HIGHER DIMENSIONS

For $n \geq 1$ let $\chi(\mathbb{Q}_p^n)$ be the chromatic number of the graph on $\mathbb{Q}_p^n$ joining v and w when $\sum_i (v_i - w_i)^2 = 1$, and $\chi_B(\mathbb{Q}_p^n)$, $\chi_m(\mathbb{Q}_p^n)$ its Borel and measurable versions.

Proposition 11. *$\chi(\mathbb{Q}_p^n)$ is finite if and only if $n = 1$, or $n = 2$ and $p \not\equiv 1 \pmod{4}$, or $p = 2$ and $n \leq 4$. Moreover:*

- (1) $\chi(\mathbb{Q}_p^1) = 2$ for every p , while $\chi_B(\mathbb{Q}_2^1) = 2$ and $\chi_B(\mathbb{Q}_p^1) = \chi_m(\mathbb{Q}_p^1) = 3$ for odd p ;
- (2) $\chi(\mathbb{Q}_2^2) = \chi(\mathbb{Q}_2^3) = 2$ and $\chi(\mathbb{Q}_2^4) = 4$, and the Borel chromatic numbers are the same.

Proof. Over $\mathbb{Q}_p$ every form in at least five variables is isotropic [20, Chapter IV, Theorem 6], and so is $x_1^2 + x_2^2 + x_3^2$ for odd p : it has a nonsingular zero modulo p by the Chevalley–Warning theorem, which lifts by Hensel’s lemma. For $p = 2$ the forms $x_1^2 + \cdots + x_n^2$ with $n \leq 4$ are anisotropic: a primitive zero $(a_1, \dots, a_4) \in \mathbb{Z}_2^4$ would have $t \geq 1$ odd entries with $t \equiv \sum_i a_i^2 \equiv 0 \pmod{4}$, so $t = 4$, and then $\sum_i a_i^2 \equiv 4 \pmod{8}$. Proposition 3 gives $\chi = \infty$ in the isotropic cases, and Corollary 4 treats $n = 2$.

(1) The neighbours of x are $x \pm 1$, so each coset of $\mathbb{Z}$ in $\mathbb{Q}_p$ is a two-way infinite path, and colouring it by the parity of the offset from a chosen representative gives $\chi(\mathbb{Q}_p^1) = 2$. For a Borel colouring use the cosets of $\mathbb{Z}_p$ instead: $x \mapsto (x - r(x)) \bmod p$, with r as in the proof of Proposition 8, maps the graph to the cycle $\mathbb{Z}/p$ with steps ± 1 , which needs 2 colours for $p = 2$ and 3 for odd p . Conversely, for odd p a measurable 2-colouring c would satisfy $c(x+2) = c(x)$; since 2 generates a dense subgroup of $\mathbb{Z}_p$, the translation by 2 is ergodic on $\mathbb{Z}_p$, so c would be almost everywhere constant on $\mathbb{Z}_p$, which contradicts $c(x+1) \neq c(x)$.

(2) The graph has edges, and $0, (1, 0, 0, 0), (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}), (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ are pairwise at distance 1, so $\chi(\mathbb{Q}_2^4) \geq 4$. For $n \leq 3$, if $\sum_i x_i^2 = 1$ with $\min_i v_2(x_i) = -k < 0$, then $\sum_i a_i^2 = 4^k$ with $a_i = 2^k x_i \in \mathbb{Z}_2$ not all even; but a sum of at most three squares of which $t \geq 1$ are odd is $\equiv t \not\equiv 0 \pmod{4}$. So unit vectors are integral, and as $\sum_i x_i^2 \equiv 1 \pmod{8}$ exactly one coordinate is odd; colouring each coset of $\mathbb{Z}_2^n$ by the parity of the coordinate sum (after subtracting its representative) gives $\chi(\mathbb{Q}_2^n) = 2$ for $n = 2, 3$. For $n = 4$ identify $\mathbb{Q}_2^4$ with Hamilton’s quaternions H over $\mathbb{Q}_2$ by $x \mapsto x_1 + x_2 i + x_3 j + x_4 k$, so that the form becomes the reduced norm N . As N is anisotropic, H is a division algebra; its valuation ring $\mathcal{O}_H = \{x : Nx \in \mathbb{Z}_2\}$ (the 2-adic Hurwitz order, $\mathbb{Z}_2^4 \cup (h + \mathbb{Z}_2^4)$ with $h = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$) has the maximal ideal $P = \{x : Nx \in 2\mathbb{Z}_2\} = \{x \in \mathbb{Z}_2^4 : \sum_i x_i \in 2\mathbb{Z}_2\}$ and the residue field $\mathcal{O}_H/P \cong \mathbb{F}_4$ [23, Proposition 13.3.4 and Theorem 13.3.11]. A unit vector lies in $\mathcal{O}_H \setminus P$, so adjacent points lie in one coset of $\mathcal{O}_H$ and their difference is nonzero in $\mathcal{O}_H/P$; fixing a representative $r(v)$ of each coset (the cosets are open) and colouring v by $(v - r(v)) \bmod P \in \mathbb{F}_4$ gives $\chi(\mathbb{Q}_2^4) \leq 4$. All these colourings are locally constant. $\square$

The values for $n = 3, 4$ are the 2-adic forms of Benda and Perles’s $\chi(\mathbb{Q}^3) = 2$ and $\chi(\mathbb{Q}^4) = 4$ [1], as cited in [13, ¶1.6] (see also [21, §§11.3–11.4]); we have not read their proofs. Since $\mathbb{Q}^n \subset \mathbb{Q}_2^n$ and the tetrahedron above is rational, the proposition proves both values again. By Corollary 4, and in one variable because the graph of ax^2 is empty or is $x \sim x \pm b$, which $x \mapsto x/b$ turns into $x \sim x \pm 1$ with a Borel colouring in at most three colours (through the residue field as in (1), or by $[\operatorname{Re} x] \bmod 2$ over $\mathbb{R}$ and $\mathbb{C}$), a non-degenerate quadratic graph over a local field of characteristic 0 never has finite chromatic number and infinite Borel chromatic number, the behaviour that [3, Remark 1.3] asks about; on the p -adic line with p odd the two differ by one, as for the graphs of irrational rotations.

7. QUESTIONS

- (1) For $p \equiv 3 \pmod{4}$, $\chi_B(\mathbb{Q}_p^2) \rightarrow \infty$, but the ordinary $\chi(\mathbb{Q}_p^2)$ is only known to be at least 4 for $7 \leq p < 2\,129\,503\,819$ (Theorem 5) and at least 5 at $p = 83$. Is it bounded? A finite subgraph whose coordinates generate a field with a real embedding lies in $\mathbb{R}^2$, so more

than seven colours would need a field with no real embedding. If $\chi(K^2)$ is unbounded over number fields K in which -1 is not a square, then so is $\chi(\mathbb{Q}_p^2)$ over $p \equiv 3 \pmod{4}$, since by Chebotarev's theorem each such K embeds in $\mathbb{Q}_p$ for infinitely many $p \equiv 3 \pmod{4}$. If $\chi(\mathbb{Q}_p^2)$ is bounded, the p -adic planes are a quantitative analogue, along a family, of the gap between ordinary and Borel chromatic numbers that [3, Remark 1.3] asks about (whose own instance, for indefinite real forms, is ruled out by Davies's theorem).

- (2) Is $\chi(\mathbb{Q}_{11}^2)$ equal to 4 or 5? Since 47, 3 and 5 are squares modulo 11, the field $\mathbb{Q}_{11}$ contains $\mathbb{Q}(\sqrt{47})$ and $\mathbb{Q}(\sqrt{3}, \sqrt{5})$; so $\chi(\mathbb{Q}_{11}^2) = 4$ would give $\chi(\mathbb{Q}(\sqrt{47})^2) = 4$, a case Moorhouse asked about [18] that is still open [11], and $\chi(\mathbb{Q}(\sqrt{3}, \sqrt{5})^2) \leq 4$. The colourings of $\mathbb{Q}_{11}^2$ that are constant on the cosets of $11^k \mathbb{Z}_{11}^2$ need five colours for $k = 1, 2, 3$ [10]. At $p = 19$ the question about $\mathbb{Q}(\sqrt{47})$ is the same, as $47 \equiv 3^2 \pmod{19}$, but $\mathbb{Q}(\sqrt{3}, \sqrt{5})$ does not embed in $\mathbb{Q}_{19}$; there we only know that the colourings constant on the cosets of $19\mathbb{Z}_{19}^2$ need five colours.
- (3) How fast does $\chi_B(\mathbb{Q}_p^2)$ grow? Theorem 1 places it between $(\frac{1}{2} + o(1))\sqrt{p}$ and $(p + 1)/2$.

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